

Supplementary Document S1 — Pre-registered Analysis

Plan for Stage 2 Triangulation

For deposition at OSF (Open Science Framework) or Zenodo with DOI

assignment Document type: Pre-specified analysis protocol Frozen timestamp:

2026-05-21 (Stage 1 results documented, Stage 2 framework specified, prior to

scRNA download and MuSiC deconvolution execution) Project: SIRT6 in

inflammatory bowel disease — single-cell, bulk, and external-validation analysis

Corresponding author: Bang Changseok (방창석) Manuscript: “Triangulated

single-cell, bulk, and external validation analyses identify SIRT6 as an epithelial-

enriched inhibitor of the inflammasome axis in inflammatory bowel disease, with

MAP3K7 (TAK1) as the most direct downstream target”

1. Purpose of this document

This document records, **before execution**, the analytical decisions taken after the Stage 1 bulk-RNA pilot analysis and before any single-cell, deconvolution, partial-correlation, or external-validation analysis was performed. Its purpose is to provide a transparent, immutable record of:

1. The Stage 1 bulk-pilot results that triggered analytical refinement.

2. The pre-specified Stage 1 go / no-go criteria and their actual outcomes.
3. The refined working hypothesis after Stage 1 (the “two-layer / three-layer” framing).
4. The pre-specified Stage 2 analytical framework, statistical thresholds, and decision rules — **set before any Stage 2 data were viewed.**

This document is archived to demonstrate that the three-layer triangulation framework was not a post-hoc construction, and that the eventual single-cell, partial-correlation, and external-validation analyses tested an already-specified hypothesis rather than being selected after the fact.

2. Stage 1 protocol (executed) and outcome

2-1. Stage 1 data and pipeline

- **Discovery cohort:** GSE193677 (Mount Sinai Crohn’s and Colitis Registry; Argmann et al. 2022, *Gastroenterology*); n = 2,490 mucosal biopsies; Illumina HiSeq 2500 bulk RNA-Seq. QC retained 2,483 samples (CD 1,154 / UC 869 / Control 460).
- **Pipeline:** DESeq2 variance-stabilizing transform → Ensembl → SYMBOL mapping (org.Hs.eg.db) → Spearman correlation (SIRT6 vs. seven targets: NLRP3, CASP1, IL1B, PYCARD (ASC), MAP3K7 (TAK1), RELA (NF-κB p65), FOXC1) across eight pre-specified subgroups → WGCNA (signed, blockwise, power = 14, mergeCutHeight = 0.25) on UC_inflamed + Control subset → GO BP / KEGG enrichment of SIRT6-containing module.

- **Target gene rationale:** TAK1, NLRP3, CASP1, IL1B, PYCARD, RELA selected as canonical NLRP3-inflammasome axis components based on published literature; FOXC1 included as a hypothesized positive-direction control based on Mahmoud et al. 2024 *Int Immunopharmacol* (PMID 38878489), which reported a Remdesivir → SIRT6 ↑ → FoxC1 ↑ → NLRP3 ↓ axis in acetic-acid-induced rat colitis.

2-2. Stage 1 pre-specified go / no-go decision rule

Pre-specified prior to viewing GSE193677 data:

GO for full follow-up paper: ≥ 3 of 7 target genes show Spearman $|\rho| \geq 0.25$ with $BH-q < 0.05$ in the UC_inflamed subgroup AND the SIRT6-containing WGCNA module shows enrichment in NLRP3-, NF- κ B-, inflammasome-, or cytokine-related Gene Ontology Biological Process terms.

NO-GO (cease project): no subgroup shows $|\rho| \geq 0.15$ with $q < 0.10$ for any of the 7 targets AND the WGCNA SIRT6 module shows no inflammation-related GO BP enrichment.

CAUTION (re-evaluate scope): any other combination — requires explicit analytical refinement before proceeding to Stage 2.

2-3. Stage 1 actual outcome (snapshot of results at time of S1 freezing)

Disclosure of analysis-order: The Stage 1 results summarized below had already been executed, and the underlying data viewed, at the time this document was frozen. The Stage 2 triangulation framework defined in §4 was

60 specified *without* any Stage 2 data (single-cell, MuSiC deconvolution, pseudobulk
61 per donor, or external validation cohort) having been viewed. This document
62 therefore serves as an immutable pre-registration of the Stage 2 analytical
63 commitments only, not of Stage 1.

Pre-specified criterion	Actual outcome
≥3 targets with strong correlation in UC_inflamed	2 of 7 met (TAK1 $\rho = -0.44$, NLRP3 $\rho = -0.39$; $q < 0.05$)
WGCNA SIRT6 module enriched for inflammation/NLRP3/NF- κ B GO BP	No — module enriched only for epithelial / barrier / wound healing terms
Direction-consistency across subgroups	TAK1 / NLRP3 consistent across Control, UC, CD, inflamed, non-inflamed
RELA correlation	Positive ($\rho = +0.25$ all samples) — opposite the canonical SIRT6 inflammasome-suppressor hypothesis
FOXC1 correlation	Weakly negative ($\rho = -0.10$) — failed to replicate Mahmoud 2024 positive coupling

64 Full Stage 1 result table archived as Appendix A
65 (pilot_analysis/results/01_sirt6_correlation_table.csv) and heatmap as Appendix
66 B (pilot_analysis/figures/02_sirt6_correlation_heatmap.png).

2-4. Stage 1 decision: CAUTION

Per the pre-specified rule, the Stage 1 outcome triggered **CAUTION**, not GO. The project is therefore obligated to undertake explicit analytical refinement before any further analyses. The remainder of this document records that refinement.

3. Working hypothesis refinement after Stage 1

3-1. Reasoning trail

1. The pre-specified criterion failed on the WGCNA inflammation-enrichment requirement, not on the correlation magnitude. SIRT6 was placed in an epithelial integrity module (hub genes HNF4A, CLDN3, CLDN4, ESRP1, ESRP2, OVOL1, KDF1, PRSS8, CES3, BDH1, HSD11B2, GPA33), and GO BP enrichment was overwhelmingly barrier-associated (sodium ion transport, establishment of skin barrier, wound healing, epidermis development, tissue morphogenesis).
2. The TAK1 and NLRP3 bulk-marginal negative correlations of $\rho \approx -0.5$ in *Control* samples ($n = 460$), where IBD-driven cell-composition shifts are absent, suggested that a substantial component of the bulk negative coupling may reflect intrinsic tissue composition (epithelial vs. immune cell relative abundance) rather than disease-induced regulation.
3. The opposite-direction RELA result (bulk $\rho = +0.25$, hypothesized direction = negative) is consistent with epithelial-program co-expression (where SIRT6 and

RELA mRNAs both track epithelial abundance), with the possibility that single-cell measurements would reveal a different relationship.

4. Bulk RNA-Seq cannot distinguish among three biologically distinct scenarios: (i) SIRT6 is regulating inflammasome genes within individual cells; (ii) SIRT6 expression and inflammasome expression occur in different cell types whose proportions covary; (iii) some combination of (i) and (ii) at different scales. **A multi-scale analytical framework is therefore required.**

3-2. Refined working hypothesis

The Stage 1 results are interpreted not as a failure of the original SIRT6-inflammasome-axis hypothesis, but as evidence that the relationship operates at multiple, possibly distinct, scales of biological organization. The refined working hypothesis is:

“SIRT6 is enriched in colonic epithelial cells and inversely couples with the NLRP3 inflammasome axis. The relative contributions of (a) tissue cell composition (epithelial vs. immune compartment), (b) within-tissue epithelial program co-expression, and (c) within-individual-cell direct molecular coupling are unknown and must be resolved by integrating analyses at three independent scales (bulk-tissue level, cell-composition-adjusted bulk level, and single-cell within-cell-type level).”

This hypothesis is broader than — and contains as a special case — the original NLRP3-suppressor hypothesis. **The refined hypothesis is not the original hypothesis with the outcome already known; it is a structural statement about scales of measurement.**

4. Pre-specified Stage 2 triangulation framework

4-1. Three-layer analytical framework (specified before execution)

Each of the seven SIRT6-inflammasome target relationships will be assessed at three independent scales:

- **Layer 1 — bulk marginal Spearman correlation** (already completed as Stage 1, GSE193677 n = 2,483).
- **Layer 2 — bulk Spearman correlation partial for the per-sample epithelial cell fraction**, where the epithelial fraction is estimated by MuSiC deconvolution using single-cell reference (see §4-3).
- **Layer 3 — within-cell-type Spearman correlation in single-cell RNA-Seq data**, computed separately within the Epithelial and Immune broad cell categories.

A target is considered to have **scale-convergent SIRT6 coupling** if it satisfies, at minimum, $|\rho| \geq 0.20$ with $q < 0.05$ in the same direction across all three layers in the UC_inflamed subgroup (or its single-cell analogue). A target is considered **cell-composition-explained** if the bulk-marginal coupling (L1) attenuates by $\geq 60\%$ in magnitude or reverses direction

after epithelial-fraction adjustment (L2). A target is considered **scale-discordant** if the bulk-level direction differs from the single-cell within-cell-type direction (suggesting Simpson's paradox).

4-2. Pre-specified statistical procedures

- **Layer 1 (re-use Stage 1):** Spearman ρ with BH-adjusted q per subgroup, as already executed.
- **Layer 2 (partial Spearman):** Rank-based partial correlation. For samples i , x_i = SIRT6 expression, y_i = target expression, z_i = MuSiC-estimated epithelial fraction. Compute r_x = residuals of $\text{lm}(\text{rank}(x) \sim \text{rank}(z))$ and r_y = residuals of $\text{lm}(\text{rank}(y) \sim \text{rank}(z))$; the partial Spearman is Pearson correlation of r_x vs r_y . P-values from cor.test on residuals, BH-adjusted per subgroup. Subgroups: All_samples, Control, UC_inflamed, CD_inflamed.
- **Layer 3 (single-cell within-cell-type Spearman):** For each broad cell type (Epithelial, Immune), compute Spearman ρ between SIRT6 and each target across all cells where at least one of (SIRT6, target) is non-zero (drop double-zero pairs to reduce dropout-driven attenuation). BH-adjusted q per broad cell type.

4-3. Pre-specified data sources for Stage 2

- **Single-cell reference:** GSE116222 (Smillie et al. 2019 *Cell*, validation cohort, 11,175 cells from 3 donors $\times$ 3 conditions [healthy, UC inflamed, UC non-inflamed]). The smaller validation cohort is selected over the larger SCP259 discovery cohort

because (a) registration-gated access for SCP259 introduces unspecified delay; (b) 11,175 cells is sufficient for broad cell-type localization of SIRT6 and within-broad-type correlation; (c) the full 366,650-cell atlas is reserved for revision-stage fine-subtype analysis if requested by reviewers.

- **Deconvolution method:** MuSiC v1+ (Wang et al. 2019 *Nat Commun*) with Smillie scRNA as reference (broad cell types as clusters, donor × condition as subjectID). CIBERSORTx (Stanford web server, S-mode batch correction) will be used as a cross-validation method for the cell fractions if MuSiC results meet criterion in §4-4.
- **Protein-level orthogonal evidence:** Human Protein Atlas SIRT6 antibody HPA071776 immunohistochemistry images for colon, rectum, duodenum, pancreas, and liver. Used to confirm or refute the predicted epithelial enrichment at the protein level.
- **External validation cohort:** GSE235236 (n = 56 sigmoid colon biopsies; 8 healthy controls, 26 UC, 22 CD; Illumina HiSeq 3000 with RSEM expected_count). The Stage 1 pipeline (DESeq2 vst → Spearman → WGCNA) will be applied verbatim to enable direct cohort-to-cohort comparison.

4-4. Pre-specified Stage 2 decision rules

The following decisions will be made on the basis of Stage 2 results, before manuscript drafting commences:

- **Primary endpoint:** Identification of at least one target with scale-convergent SIRT6 coupling (per §4-1) AND direction-consistent reproduction in the GSE235236 external cohort. If achieved, the target becomes the manuscript's central finding.
- **Methodological control (RELA paradox):** If the bulk RELA-SIRT6 positive correlation is reversed to negative within single cells (suggesting Simpson's paradox), pseudobulk-per-donor analysis must reproduce the within-cell-type negative direction to exclude scRNA dropout as the explanation. Pseudobulk groups will be donor × condition × broad cell type with ≥ 10 cells; raw counts summed and DESeq2 vst applied.
- **Cell-composition cross-validation:** MuSiC cell fractions will be cross-validated against CIBERSORTx S-mode estimates. Reasonable agreement (Spearman $r \geq 0.7$ across samples for the dominant cell type) is required for partial-correlation results to be reported as the main finding.
- **External cohort threshold:** At least 60% of 28 discovery-validation comparison pairs (7 targets × 4 subgroups) must show direction consistency (sign agreement) for the discovery findings to be considered externally reproducible.

4-5. Multiple testing handling (pre-specified)

Within each scale (each subgroup-target combination within Layer 1, Layer 2, Layer 3, or the validation cohort), Benjamini-Hochberg false discovery rate correction is applied per subgroup. **No additional cross-layer correction is applied**, following recommended practice for triangulation designs (Lawlor et al. 2016 *International Journal of Epidemiology*):

triangulation is convergent evidence across independent analytical scales, not multiple hypothesis testing of the same scale. This decision is documented prior to Stage 2 execution and is not revisable after results are observed.

4-6. What this document does NOT pre-specify

The following aspects are explicitly NOT specified in this document and are subject to data-driven choice at execution time:

- Specific scRNA clustering resolution (Seurat FindClusters resolution parameter)
- Specific cell-type annotation method (SingleR reference set; the Smillie original 51-cell-subtype labels would have been preferred but require SCP259 registration)
- Specific WGCNA module identification on the external validation cohort (sample size is small; module reproduction is not the primary endpoint)
- The decision of which downstream target (TAK1, NLRP3, RELA, or other) emerges as the primary scale-convergent finding — this is the actual scientific question and is by construction unknown in advance

5. Outcomes to be reported

Per the framework above, the following are pre-specified deliverables of Stage 2:

1. **A single-cell expression localization figure** (UMAP + dot plot + violin) showing the SIRT6 cell-type distribution in human colonic mucosa, with broad cell-type

annotation. Whether SIRT6 is preferentially epithelial or immune is the actual scientific question and is unknown in advance.

2. **A within-cell-type correlation heatmap** (Layer 3) showing SIRT6 vs. seven targets within Epithelial and Immune categories.
3. **A bulk marginal vs. partial correlation figure** (Layer 1 vs. Layer 2) showing which bulk correlations are robust to cell-composition adjustment and which attenuate or reverse.
4. **An external validation figure** showing Spearman ρ from GSE235236 with direction-consistency vs. GSE193677.
5. **A pseudobulk-per-donor figure** confirming or refuting the dropout-artifact alternative for the RELA paradox.
6. **A transparent reporting of the Stage 1 CAUTION outcome and the refinement reasoning**, in the Methods section, with reference to this document's DOI.

6. Archive details

- **Author of record:** Bang Changseok
- **Project identifier:** SIRT6-IBD-2026
- **Freeze date:** 2026-05-21 (post Stage 1 completion, pre Stage 2 execution)
- **Deposition:** OSF (Open Science Framework), DOI: 10.17605/OSF.IO/AB5MY

- **Citation in manuscript:** “The Stage 2 triangulation framework was pre-specified at DOI: 10.17605/OSF.IO/AB5MY prior to scRNA-Seq, MuSiC deconvolution, and external-validation analyses.”
- **Version control:** This document is the v1.0 freeze. Any modifications after Stage 2 execution must be appended as a separate “post-hoc amendment” record with their own timestamp.

7. Appendices (referenced)

Appendix A — Stage 1 SIRT6 vs target correlation table

File: pilot_analysis/results/01_sirt6_correlation_table.csv (already generated prior to S1 freezing). All 8 subgroups × 7 targets × {p, p, q}.

Appendix B — Stage 1 SIRT6 correlation heatmap

File: pilot_analysis/figures/02_sirt6_correlation_heatmap.png.

Appendix C — Stage 1 WGCNA SIRT6 module hub gene list

File: pilot_analysis/results/02_sirt6_module_hub_top30.csv. 30 hub genes ranked by intramodular kME, including HNF4A, GPA33, CLDN3, CLDN4, ESRP1, ESRP2, OVOL1, KDF1, PRSS8, CES3, BDH1, HSD11B2.

239 Appendix D — Stage 1 GO BP enrichment of SIRT6 module

240 File: pilot_analysis/results/03_GO_BP_sirt6_module.csv. Top 10 terms include
241 sodium ion transport, establishment of skin barrier, sodium ion transmembrane transport,
242 wound healing, mechanoreceptor differentiation, embryonic morphogenesis, epidermis
243 development. **No NLRP3-, NF-κB-, inflammasome-, or cytokine-related term passes $q <$**
244 **0.05.**

245 Appendix E — Original go / no-go decision framework

246 File: pilot_analysis/go_nogo_decision_framework.md. Frozen prior to Stage 1 execution;
247 the rules in §2-2 above are reproduced verbatim from this file.

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249 *End of S1 archive document. This file is the immutable historical record of the analytical*
250 *framework that governed Stage 2 of the SIRT6-IBD project. Any deviation from this*
251 *framework during execution is reported transparently in the manuscript Methods and as a*
252 *post-hoc amendment record.*